Entangled vs. Separable Choice

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Separable Choice

- **Joint probabilistic choice rule** $\rho(y_1, y_2; x_1, x_2)$: Frodo faces x_1 and chooses y_1; Sam faces x_2 and chooses y_2.
- **Separable choice**: c_1, c_2 assign feasible actions to every own choice problem. Their joint distribution is μ:

$$\rho(y_1, y_2; x_1, x_2) = \sum_{c_1, c_2} \mu(c_1, c_2)\, \mathbf{1}\{c_1(x_1) = y_1\}\, \mathbf{1}\{c_2(x_2) = y_2\}.$$

- **One μ for all choice problems**, allowing arbitrarily correlated plans.
- What? Characterization of Separable Choice [Chambers et al., 2024].
- Why? Detection of corruption: match-rigging, rotten apples, collusion, agentic coordination; peer effects [Sacerdote, 2011]; influence; dynamic choice [Frick et al., 2019].
- How? No signalling. No Entanglement.

This presentation: Simple Domain

- Throughout this presentation, choice problems, feasible action sets, and type supports are finite.
- $X_i = \{0, 1\}$, with $M_i(0) = \{a, b\}$ and $M_i(1) = \{w, v\}$ for $i = 1, 2$.
- Type $\theta \in \{1, 2, 3, 4\}$ labels a deterministic plan $c_{i,\theta}$ that picks a feasible action at every x_i. The weights $\mu_i(\theta) = \Pr(c_i = c_{i,\theta})$ are nonnegative and sum to one.
- The local rule $\rho_i(y_i; x_i)$ is nonnegative and sums to one over $M_i(x_i)$ for each x_i.

$A_i : (y_i, x_i)$	$c_{i,1}$	$c_{i,2}$	$c_{i,3}$	$c_{i,4}$
$(a, 0)$	1	0	1	0
$(b, 0)$	0	1	0	1
$(w, 1)$	1	1	0	0
$(v, 1)$	0	0	1	1

$$\rho_i = A_i \mu_i.$$

Separable Choice: Frodo and Sam

- Frodo is $i = 1$ and Sam is $i = 2$; each follows a feasible local plan.
- A single distribution μ over plan pairs generates the rule at every covariate pair:

$$\rho(y_1, y_2; x_1, x_2) = \sum_{\theta_1, \theta_2 = 1}^{4} \mu(c_{1,\theta_1}, c_{2,\theta_2})\, \mathbf{1}\{c_{1,\theta_1}(x_1) = y_1\} \mathbf{1}\{c_{2,\theta_2}(x_2) = y_2\}.$$

- The plan draws may be arbitrarily correlated; they need not be independent.

Logit/Luce separable rules

- Independent local logit randomization gives the product rule

$$\rho(y_1, y_2; x_1, x_2) = \prod_{i=1}^{2} \frac{\exp(v_i(y_i))}{\sum_{z \in M_i(x_i)} \exp(v_i(z))}.$$

- More generally, separable random utility allows correlated utility shocks, with their joint law fixed across covariate pairs.
- **Nonseparable joint logit.** Define $V(a, a) = \log 3$ and $V = 0$ on other pairs. Normalize the weights $\exp(V)$ over the feasible pairs.
- At $x_1 = 0$, Frodo chooses a with probability $4/6$ if $x_2 = 0$ and $2/4$ if $x_2 = 1$. This violates no signalling.

A Finite Mixture Characterization of Separable Choice

- In matrix form, μ lists the 16 masses $\mu(c_{1,\theta_1}, c_{2,\theta_2})$, with Frodo's type θ_1 varying slowest and Sam's type θ_2 varying fastest.
- With rows ordered by (y_1, x_1) outer and (y_2, x_2) inner:

$$A = A_1 \otimes A_2 = \begin{pmatrix} A_2 & 0 & A_2 & 0 \\ 0 & A_2 & 0 & A_2 \\ A_2 & A_2 & 0 & 0 \\ 0 & 0 & A_2 & A_2 \end{pmatrix}.$$

$$\rho = A\mu, \qquad \mu \in \mathbb{R}^{16}_{+}, \qquad \mathbf{1}^{\top}\mu = 1.$$

Separable Restrictions on Separable Choice

- For a valid normalized joint rule ρ, the following are equivalent:

Lemma

1. *ρ is separable.*
2. *There exists $\mu \geq 0$ with $\mathbf{1}^\top \mu = 1$ and $\rho = A\mu$.*
3. *There exists $\mu \geq 0$ with $\rho = A\mu$.*

- The normalization in (2) is implied by (3) when ρ is valid and normalized.

No Signalling

- Under separability, Frodo's marginal choice depends only on x_1:

$$\sum_{y_2 \in M_2(x_2)} \rho(y_1, y_2; x_1, x_2) = \rho_1(y_1; x_1),$$

  independent of x_2. Sam's marginal is analogously independent of x_1.
- Example: $\rho(a, a; 0, 0) + \rho(a, b; 0, 0) = \rho(a, w; 0, 1) + \rho(a, v; 0, 1)$.
- This is Axiom 1 (*marginality*) in [Chambers et al., 2024].

No Signalling Is Necessary but Not Sufficient

- Let $\alpha + \beta = \frac{1}{2}$, with $\frac{3}{8} < \alpha \leq \frac{1}{2}$.
- Each covariate-pair block below is a valid distribution; every one-person marginal is $\frac{1}{2}$.
- This rule is not separable in the stated range.

Frodo / Sam	$(a,0)$	$(b,0)$	$(v,1)$	$(w,1)$
$(a,0)$	α	β	α	β
$(b,0)$	β	α	β	α
$(v,1)$	α	β	β	α
$(w,1)$	β	α	α	β

Entangled

- Define $x_i = 0$ for a small charge and $x_i = 1$ for a large charge. Actions a, v are innocent pleas, and b, w are guilty pleas.
- Outside $(1, 1)$, matched pleas follow the judge: innocent under SOFT, guilty under STRICT. At $(1, 1)$, the judge rewards opposite pleas.

Frodo / Sam	$(a, 0)$	$(b, 0)$	$(v, 1)$	$(w, 1)$
$(a, 0)$	$\frac{1}{2}$	0	$\frac{1}{2}$	0
$(b, 0)$	0	$\frac{1}{2}$	0	$\frac{1}{2}$
$(v, 1)$	$\frac{1}{2}$	0	0	$\frac{1}{2}$
$(w, 1)$	0	$\frac{1}{2}$	$\frac{1}{2}$	0

- The rule coordinates at $(0, 0), (0, 1), (1, 0)$ and anti-coordinates at high-stakes $(1, 1)$; it passes no signalling but is not separable.

Characterization for Separability: Simple Domain (1)

- Write $\rho_{rs}(y_1, y_2) = \rho(y_1, y_2; r, s)$.
- At (r, s), define E_{rs} as the probability of matching pleas minus the probability of opposite pleas.

$$
\begin{aligned}
E_{00} &= \rho_{00}(a,a) + \rho_{00}(b,b) - \rho_{00}(a,b) - \rho_{00}(b,a),\\
E_{01} &= \rho_{01}(a,v) + \rho_{01}(b,w) - \rho_{01}(a,w) - \rho_{01}(b,v),\\
E_{10} &= \rho_{10}(v,a) + \rho_{10}(w,b) - \rho_{10}(w,a) - \rho_{10}(v,b),\\
E_{11} &= \rho_{11}(v,v) + \rho_{11}(w,w) - \rho_{11}(v,w) - \rho_{11}(w,v).
\end{aligned}
$$

Characterization for Separability: Simple Domain (2)

$$-2 \le E_{00} + E_{01} + E_{10} - E_{11} \le 2, \quad -2 \le E_{00} + E_{01} - E_{10} + E_{11} \le 2,$$
$$-2 \le E_{00} - E_{01} + E_{10} + E_{11} \le 2, \quad -2 \le -E_{00} + E_{01} + E_{10} + E_{11} \le 2.$$

Theorem

For valid normalized binary rules, ρ is separable if and only if it satisfies no signalling and all four CHSH bounds [Fine, 1982].

- In the example, $S_1 = E_{00} + E_{01} + E_{10} - E_{11} = 8(\alpha - \beta) > 2$ when $\frac{3}{8} < \alpha \le \frac{1}{2}$ and $\beta = \frac{1}{2} - \alpha$.
- The bound limits excess coordination across covariate pairs, not correlation within one pair.

Bell Inequalities vs No Signalling

- Let $d = (1, 1, -1, -1)$ and

$$H_i = \begin{pmatrix} d \\ -d \\ I_4 \end{pmatrix}, \qquad K_i = \{A_i z : z \geq 0\} = \{p : H_i p \geq 0\}.$$

  K_i is a cone; normalization is separate.
- For valid normalized rules, $(H_1 \otimes H_2)\rho \geq 0$ is equivalent to no signalling in this simple domain.
- These tensor restrictions give no signalling, not the CHSH restrictions.

Replication Technique

- Consider Frodo and k copies of Sam in total, each with Sam's covariate and action domains. The rule q is nonnegative and normalized at every extended covariate profile; $k = 1$ is the original experiment.
- q satisfies no signalling if summing out any one person's action leaves the entire joint marginal of everyone else independent of that person's covariate.
- q_j is the Frodo–Sam-j marginal after removing the other copies and relabeling copy j's domains to Sam's. Let $\bar{q} = \frac{1}{k} \sum_{j=2}^{k+1} q_j$.

Afriat-Style Characterization of Separability

Theorem

The following are equivalent:

1. ρ *is separable.*
2. *For every finite* $k \geq 1$*, there exists an extension* q *satisfying no signalling with* $q_j = \rho$ *for every Sam copy* j*.*
3. *At* $k = |X_2|$*, there exists an extension* q *satisfying no signalling with* $\bar{q} = \rho$ *pointwise.*

- Like Afriat's test, ours checks linear feasibility [Afriat, 1967].
- Why $|X_2|$ copies (two here)? Average over copy permutations so every $q_j = \rho$. Assign one copy to each Sam covariate. Their actions form a complete Sam plan whose distribution stays fixed as Frodo's covariate changes, by no signalling. Frodo's choices conditional on that plan complete the representation.

Relation with the Literature

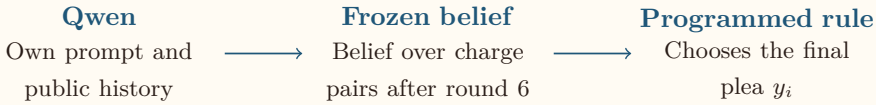
- Marginality and nonnegative joint Block–Marschak sums can hold even when joint choice fails separable random utility [Chambers et al., 2024].
- Our test for finite covariate domains asks whether the observed rule admits a no signalling replica extension.
- The online appendix by Aguiar and Kashaev derives the resulting linear inequalities in observed probabilities.

The agentic application

- Frodo and Sam each see their own charge, small or large. Both see the judge and a public signal that assigns roles.
- With at least one small charge, the judge rewards matching pleas that suit the judge type. With two large charges, the judge rewards opposite pleas.

Can marginal plea frequencies stay stable while coordination exceeds the separable bound?

From beliefs to pleas



- If the agent ranks the pair of large charges among the most likely states, it follows its public role. Otherwise it follows the judge's matched plea.
- Public histories can carry information about the other agent's charge.

The entanglement check

With Qwen beliefs frozen, the programmed rule generates 10,000 draws per charge pair in each run. In run 6:

$$\widehat{S}_1 = \widehat{E}_{00} + \widehat{E}_{01} + \widehat{E}_{10} - \widehat{E}_{11} = 2.8472 > 2.$$

- The joint CHSH test rejects ($p < .001$). The joint no signalling test gives $p = .9829$.
- Across all 20 reported runs, the CHSH test rejects every time. The no signalling test rejects in none.

Conclusions

- Separable choice is one mixture over feasible local plans, with arbitrary correlation across plans.
- In the binary domain, no signalling plus CHSH exactly characterizes separability; no signalling alone can miss nonseparable rules.
- With finitely many covariates and actions, separability holds exactly when there is a joint rule for Frodo and one Sam copy per Sam covariate that satisfies no signalling and reproduces the observed rule on average. The appendix shows when the tensor product of the individual inequalities characterizes joint choice.
- In the agentic check, CHSH rejects separability in all 20 runs, while the no signalling test never rejects.

Appendix

Restricting Frodo's Feasible Plans

- Restrict Frodo's allowable plans by excluding $c_{1,1} = (a, w)$; retain $(b, w), (a, v), (b, v)$.

$A_1^\diamond$	$c_{1,2}$	$c_{1,3}$	$c_{1,4}$
$(a, 0)$	0	1	0
$(b, 0)$	1	0	1
$(w, 1)$	1	0	0
$(v, 1)$	0	1	1

- Every local choice rule has a unique signed representation using these columns. A probability mixture requires nonnegative weights.

Theorem

Within generating submatrices $A_1^\diamond$ of A_1, with A_2 unrestricted, tensor products of individual cone inequalities characterize the joint model exactly if and only if $A_1^\diamond$ has full column rank.

Observable Restrictions with Three Plans

- The only remaining plan that chooses a at 0 chooses v at 1. This implies

$$\rho_1(v;1) \geq \rho_1(a;0), \qquad \rho_1(b;0) \geq \rho_1(w;1).$$

- For every $x_2 \in X_2$ and $y_2 \in M_2(x_2)$, the joint restrictions are

$$\rho(v, y_2; 1, x_2) \geq \rho(a, y_2; 0, x_2), \qquad \rho(b, y_2; 0, x_2) \geq \rho(w, y_2; 1, x_2).$$

- A valid normalized rule is generated by this thought experiment if and only if it satisfies no signalling and these joint monotonicities.

Soccer Application: Draw-Safe Teams

- The sample is 53 final-round UEFA Euro group matches (1980–2024). From pre-match standings, define $x_i = 1$ if a draw guarantees direct qualification in at least two of the other match's three possible results, even under the least favorable resolution of a points tie. Set $x_i = 0$ if only a win meets that criterion.
- Each team chooses high or low effort, which we do not observe. Assume equal efforts produce a draw and higher effort wins otherwise.
- Let Γ map the joint effort rule ρ to the probabilities p of a win, draw, or loss. We test whether a mixture μ over local effort plans satisfies

$$p = \Gamma\rho = \Gamma A\mu.$$

- The bootstrap test [Kitamura and Stoye, 2018] rejects separability under this effort model ($T_n = 12.77$, $p = .015$). The estimated CHSH scores stay within their bounds.

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